Facades

User UI

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UserSystem

-users ArrayList<user>
-reliefRequests ArrayList<ReliefRequest>
-shelters ArrayList<Shelter>
-volunteers ArrayList<Volunteer>
-currentUser: User
-currentHurricane: Hurricane
-hurricaneTracker: Hurricane

+ Login(String userName String password): boolean
+ LogOut() :void
+SignUp(String firstName, String lastName, Date birthDate, String password): boolean
+getReliefRequests(): ArrayList<ReliefRequest>
+getHurricane(Location location): Hurricane
+getShelters(): ArrayList<Shelters>
+createReliefRequest(): ReliefRequest
+submitReliefRequest (ReliefRequest request): void
+getReliefRequest (): ArrayList <RelefRequest>
+editReliefRequest(...): void
+ getUsers(): ArrayList<User>
+ getShelters(): ArrayList<Shelter>
+ getShelterManagers(Shelter shelter): ArrayList<ShelterManager>
+assignUserToShelter (Shelter shelter, User user): boolean
+getVolunteers(): ArrayList <Volunteers>
+acceptReliefRequest(Volunteer volunteer, ReliefRequest request(): void

Database

DataConstants

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DataLoader

getUsers(): ArrayList<User>
getShelters(): ArrayList<Shelter>
getReliefRequests(): ArrayList<ReliefRequest>

DataWriter

saveUsers(): boolean
saveShelters(): boolean
saveReliefRequests(): boolean

Singletons/Lists

HurricaneList

hurricanes ArrayList<Hurricane>

+addHurricane(String hurricaneName, Location location, int category, double windSpeed, double diameter): void
+updateHurricane(UUID id, Location newLocation, int newCategory, double newWindSpeed, double newDiameter): void
+ removeHurricane(UUID id): void

ShelterList

-shelterList: Shelterlist
-shelters: ArrayList<Shelter>

+ShelterList()
+getInstance(): ShelterList
+getShelter(String shelterName, Location location): User
+addUser(UUID Id): User

UserList

-userList: UserList
-users: ArrayList<User>

+UserList()
+getInstance(): UserLlist
+getUser(String userName, String password): User
+addUser(String userName, String password, String email): boolean
+saveUser(): boolean

ReliefRequestList

-userList: UserList
-users: ArrayList<User>

+ReliefList()
+getInstance(): ReliefList
+getReliefRequest()

Hurricane

-hurricaneName: String
-location: Location
-category: int
-windSpeed: double
- diameter: double
-id: UUID

+setLocation(Location location): void
+setCategory(Category: int): void

<<enumeration>> Filter

PET FRIENDLY
FOOD
WATER
MEDICAL
OPEN CAPACITY

Shelter

Shelter

-shelterName: String
-filters ArrayList<filter>
-shelterManagersArrayList<ShelterManager>
-currentCapacity: int
-residents: List<Person>
-location: Location
-id: UUID

+setCapacity(capacity: int): void
+getAvailableSpaces(): int

Location

-zipCode: int

+getZip(): int

ShelterManager

- shelterVolunteers: ArrayList<Volunteer>

+addShelterVolunteer(Volunteer volunteer)
+removeShelterVolunteer(Volunteer volunteer)
+getShelterVolunteers<sheltervolunteers>
+updateShelterCapacity(Shelter shelter, int capcity): void

User

-firstName: String
-lastName: String
-birthDate: Date
-email: String
-password: String
-userName: String
-location: Location
-id: UUID
-requests: ArrayList<ReliefRequest>

+User(String firstName, String lastName, Date birthDate, String email, String password, String username)
+isMatch(String userName, String password): bool
+String getFirstName()
+String getLastName()
+Date getBirthDate()
+String getEmail()
+String getPassword()
+String getUsername()

ReliefRequest

-victims ArrayList<Victim>
-hazards: ArrayList<Hazard>
-photo: String
-assistanceTypeNeeded ArrayList<AssistanceType>
-status: Status
-description: String
-phoneNumber: String
-id: UUID
-location: Location

+createRequest(Location location, String description, String phoneNumber, Hazard hazards, Victim victims, String photo, AssistanctType assistanceType):
+addVictim(victim Victim): void
+removeVictim(victim Victim): void
+addAssistanceType(type AssistanceType): void
+removeAssistanceType(type AssistanceType): void
+setPhoto(photo String): void
+setStatus(status Status): void
+getStatus(): Status
+addDescription(String description): void

<<enumeration>> Status

AWAITING_ASSISTANCE
IN_PROGRESS
COMPLETE

<<enumeration>> Assistance Type

MEDICAL
TRANSPORTATION
VETINARY
DEBRIS_REMOVAL

<<enumeration>> Hazard

FLOODING
FIRE
DOWNED_POWER_LINES
GAS_LEAK
DEBRIS
STRUCTURAL_DAMAGE
ROAD_BLOCKAGE

Volunteer

Volunteer

-tools ArrayList<tool>
-vehicles ArrayList<vehicle>
-certifications ArrayList<Certification>

+addTool(Tool tool) : void
+removeTool(Tool tool) : void
+addVehicle(Vehicle vehicle) : void
+removeVehicle(Vehicle vehicle) : void
+addCertification(Certifications certification) : void
+acceptReliefRequest(ReliefRequest request) : void

<<enumeration>> tool

CHAINSAW
SAW
HAMMER
WRENCH
FLASHLIGHT
AXE
RADIO

<<enumeration>> vehicle

TRUCK
CAR
BOAT
AIR SUPPORT

<<enumeration>> Certifications

CPR
AED
PFA (Psychological First Aid)
CERT
NRP

Victim

Victim

-firstName: String
-lastName: String
-description: String
-gender: String
-age: int
-victimStatus: enum Victim Status

+Victim(firstname: String, victimStatus; VictimStatus)
+getFirstName(): String
+getLastName(): String
+getDescription(): String
+getGender(): String
+getAge(): int
+getVictimStatus();
+setVictimStatus(status: VictimStatus): void

<<enumeration>> VictimStatus

LOST
MINOR_INJURY
MAJOR_INJURY
TRAPPED
UNRESPONSIVE
DECEASED

Pet

-species: String

+Pet(VictimStatus victimStatus, species String)